```
PARTITIONS_READY first=0 last=20002 count=20003
PASS independent_partition_product n=0..20002 count=20003
PASS R0=d1(72)/d1(71)=695578/609237
VALUE R0=695578/609237
PASS q_2(R)=(R-1)^2+2>0 for every real R
BLOCK_PASS D(t,0) t=72..1071 inequalities=1000 failures=0
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BLOCK_PASS D(t,0) t=5072..6071 inequalities=1000 failures=0
BLOCK_PASS D(t,0) t=6072..7071 inequalities=1000 failures=0
BLOCK_PASS D(t,0) t=7072..8071 inequalities=1000 failures=0
BLOCK_PASS D(t,0) t=8072..9071 inequalities=1000 failures=0
BLOCK_PASS D(t,0) t=9072..9999 inequalities=928 failures=0
PASS D(t,0)>0 for all 72<=t<10000; cases=9928
PASS direct and expanded D(t,0) agree at all 9928 indices
MINIMUM D(t,0)=27986845 index=73 ties=1
BLOCK_PASS R0 s=3..34 cases=32 inequalities=64 failures=0
PASS R0: positive denominators, R>=1, both strict inequalities
MINIMUM R0_cleared_margin1=522896 index=3 ties=1
MINIMUM R0_cleared_margin2=168215963182 index=4 ties=1
BLOCK_PASS Rs s=35..1034 cases=1000 inequalities=2000 failures=0
BLOCK_PASS Rs s=1035..2034 cases=1000 inequalities=2000 failures=0
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BLOCK_PASS Rs s=6035..7034 cases=1000 inequalities=2000 failures=0
BLOCK_PASS Rs s=7035..8034 cases=1000 inequalities=2000 failures=0
BLOCK_PASS Rs s=8035..9034 cases=1000 inequalities=2000 failures=0
BLOCK_PASS Rs s=9035..9999 cases=965 inequalities=1930 failures=0
PASS Rs: positive denominators, R>=1, both strict inequalities
MINIMUM Rs_cleared_margin1=774894329 index=35 ties=1
MINIMUM Rs_cleared_margin2=149021112963685 index=35 ties=1
PASS ratio ranges cover every integer s=3..9999 exactly once
PASS core inequality count = 29922
COMPLETE_PASS core_inequalities=29922 identity_checks=9928 group_checks=9 failed=0 elapsed_us=3797774
```